

Age-related cognitive disorders and cancer: exploration of potential transcriptomic links

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Introduction

Cognitive decline in older adults is a result of natural aging, and can often be accelerated by age-related cognitive disorders (ACDs) [1]. ACDs are a collection of symptoms involving a decline in memory and thinking, involving dementia; ACDs are extremely prevalent in older adults, with 1 in 9 having some form of subjective cognitive decline [2]. While current research is sparse, some studies have explored the links between ACDs and cancer [3]. To understand the mechanistic underpinnings of ACDs, in my research I aimed to find differences at the transcriptomic level between ACD patients and controls. My results indicate the key role of these genes in modulating apoptosis and inflammation in pancreatic cancer.

Methods

I took samples from the Gene Expression Omnibus (GEO, accession number: GSE239282; accessed March 2024, from a study done by Gómez-Carballa et. al [4]), which contained capillary blood samples from ACD (n=15) and control (n=14) patients over two timepoints. I used GEO2R to analyze the differential expressed genes (DEGs) between ACD patients (14 usable samples) and control patients (12 usable samples) at time point 1(T1). Enrichment testing was done using STRING-db to find biological and functional pathways enriched for the significant genes.

Results

I found that the interactions of natural killer cells in the pancreatic cancer pathway were enriched in ACD patients at T1 (count in network = 7 of 27, false discovery rate = 1.76×10^{-5}). In that pathway, patients with ACD had significantly lower levels of transcription factor forkhead box O1 (FOXO1, adjusted p-value = 0.016, log2FoldChange = -0.406), and significantly higher levels of inflammatory genes such as CCL4 (adjusted p-value = 0.0002, log2FoldChange = 0.986). FOXO1 is a key gene that regulates apoptosis in pancreatic cancer cells [5]. There were also significantly lower levels of the pro-apoptotic WT1 regulator (PAWR, adjusted p-value = 0.005, log2FoldChange = 0.855), a gene that induces apoptosis in cancer cells, as compared to control patients.

Discussion

My results suggest that the onset of ACDs may be linked to a decline in the body's ability to fight pancreatic cancer. The reduced presence of key genes FOXO1 and PAWR, both of which are major regulators of cancer cell apoptosis, as well as the increased presence of inflammatory genes such as CCL4 that excite the immune system, could result in inefficiencies in immune responses to cancer in ACD patients.

However, my research is contradictory to existing literature that states that susceptibility to Alzheimer's is inversely linked with susceptibility to cancer [6]. In addition, the small sample size is a limitation, and there is a confounding variable in that the average age of the ACD group is much higher than the control group. Further research will be needed to properly investigate this link, given the multitude of genes involved in this process. Future research could focus on eliminating confounding variables, larger sample size studies, and further validation through biological experiments.

Keywords: Alzheimer's disease, age-related cognitive disorders, pancreatic cancers, inflammatory responses, apoptosis

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